



2134 - Absolute Brightness Measuremt of the Extragalactic Background Light Using Gallilean Satellites Eclipse Occultations

Cycle: 1, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Europa eclipse				
	1	Europa eclipse	NIRCam Imaging	(1) EUROPA-OFFSET
	2	Europa eclipse	NIRCam Imaging	(2) EUROPA-OFFSET2
Ganymede Eclipse				
	3	Ganymede eclipse	NIRCam Imaging	(5) GANYMEDE-OFFSET-V3
	4	Ganymede eclipse	NIRCam Imaging	(6) GANYMEDE-OFFSET-V4

ABSTRACT

The Extragalactic Background Light (EBL) as an integrated history of the early universe is important for the study of unresolved star formation. However, previous EBL measurements suffer from residual contamination from strong foregrounds, the zodiacal light (ZL). We propose to observe Galilean satellites eclipsed in the Jovian shadow as occulters to detect the absolute EBL intensity without any ZL uncertainty. ZL originates inside

the Jovian orbit; since the Galilean satellites in eclipse shield all light beyond the Jovian orbit, they should be detected as 'dark spots' if the strong EBL implied by previous observations exists. The intensity deficit of this dark spot relative to the surrounding sky directly measures the brightness of EBL, free from any assumptions about ZL. The observation condition for EBL is perfect at JWST Cycle-1 season because Jupiter locates at high Galactic latitude ($|b| > 40$ deg) and deep eclipses will occur, which is the opportunity once in 6 years. Therefore, observations in this Cycle-1 are highly required.

OBSERVING DESCRIPTION

Our observational targets are the Galilean satellites (Europa, Ganymede and Callisto), which are non-stationary targets. Since our method relies on specific astronomical events, chances for observation are limited and timing is absolutely critical; in particular, observations must begin at the specified time. The basic information of eclipse (time and position of Jupiter and the Galilean satellites) are provided by the Natural Satellites Ephemeride Server MULTISAT (<http://www.sai.msu.ru/neb/nss/nssphe0he.htm>) and the JPL-HORIZONS System (<http://ssd.jpl.nasa.gov/horizons.cgi>). For example, in a Ganymede eclipse case on Jun. 2nd, 2022, the diameter of Ganymede is 1.30 arcsec circle, it is located at 4-39 arcsec from Jovian limb, and its relative speed to the Jupiter is 2.6 mas/sec.

Our strategy is to measure the EBL brightness as a dark spot, and the zodiacal light brightness can be evaluated locally around the dark spot itself. The estimated EBL brightness is 10 nW/m²/sr at 1.4 μ m, which can be detected by NIRcam imaging with F140M filter by 1 hour integration with S/N=10. The greatest difficulty in this observation results from scattered light due to the close proximity of Jupiter to the target satellites, which will reduce S/N. However, Jupiter is dark at F140M owing to the methane absorption in the Jovian atmosphere, thus stray light from Jupiter is also reduced. In addition, we will keep Jupiter out of the detector field of view (FoV) during the observation to avoid stray light from Jupiter. Observations will be conducted with non-sidereal tracking on Jupiter (outside of the FoV) to fix the stray light pattern on the detector during the observation. This will minimize the systematic error of the Jovian stray light. Since the relative speed of the target satellites to Jupiter is ~ 2.6 mas/sec, short integrations (10 sec) are required to avoid smearing of the target satellite. Letting the eclipsed satellite move relative to the detector using Jupiter tracking has the advantage of effectively dithering the observations so that we can average out any detector issues. We emphasize that we know the ephemeris of Galilean satellites with great precision; the size and location of the Galilean satellites will be accurately known, and therefore easily differentiated from detector effects. In addition, simultaneous observations at F445W trace the position of the satellite in eclipse because thermal emission from 120 K surface of the satellite at >4 μ m is detectable even in eclipse.

Proposal 2134 - Targets - Absolute Brightness Measuremt of the Extragalactic Background Light Using Galilean Satellites Eclipse Oc...

Solar System Targets	#	Name	Level 1	Level 2	Level 3
	(1)	EUROPA-OFFSET	STD=JUPITER	TYPE=POS_ANGLE,RAD=102.5,ANG=62,REF=NO RTH	
	<i>Comments: Extended=YES</i>				
	(2)	EUROPA-OFFSET2	STD=JUPITER	TYPE=POS_ANGLE,RAD=100,ANG=250,REF=NO RTH	
	<i>Comments: Extended=YES</i>				
	(5)	GANYMEDE-OFFSET-V3	STD=JUPITER	TYPE=POS_ANGLE,RAD=90,ANG=234,REF=NOR TH	
	<i>Comments: Target for Obs 3. Extended=Unknown</i>				
	(6)	GANYMEDE-OFFSET-V4	STD=JUPITER	TYPE=POS_ANGLE,RAD=105,ANG=258,REF=NO RTH	
	<i>Comments: Target for Observation 4 Extended=Unknown</i>				

Proposal 2134 - Observation 1 - Absolute Brightness Measuremt of the Extragalactic Background Light Using Gallilean Satellites Eclip...

Observation	Proposal 2134, Observation 1: Europa eclipse										Wed Jun 24 16:00:09 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCam Imaging										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Europa eclipse (Obs 1)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.										
Solar System Targets	#	Name	Level 1			Level 2			Level 3		
	(1)	EUROPA-OFFSET	STD=JUPITER			TYPE=POS_ANGLE,RAD=102.5,ANG=62,REF=NO RTH					
	Comments: Extended=YES										
Template	Module					Subarray					
	B					FULL					
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	NONE				STANDARD				2	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F140M	F356W	SHALLOW4	5	2	4	2	1052.203		
	2	F140M	F356W	SHALLOW4	5	2	4	2	1052.203		
	3	F140M	F356W	SHALLOW4	5	2	4	2	1052.203		
	4	F140M	F356W	SHALLOW4	5	2	4	2	1052.203		
Special Requirements	ECLIPSE UMBRAL FULL OF EUROPA BY JUPITER FROM JWST										
	DEFAULT WINDOW: NOT OCCULTATION OF EUROPA-OFFSET BY JUPITER FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF EUROPA-OFFSET BY IO FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF EUROPA-OFFSET BY EUROPA FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF EUROPA-OFFSET BY GANYMEDE FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF EUROPA-OFFSET BY CALLISTO FROM JWST DEFAULT WINDOW: SEPARATION OF EUROPA-OFFSET IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA-OFFSET GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA-OFFSET CALLISTO FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE EUROPA-OFFSET FROM JWST LESS THAN 0.03 SEPARATION OF JUPITER EUROPA FROM JWST GREATER THAN 4"										

Proposal 2134 - Observation 2 - Absolute Brightness Measuremt of the Extragalactic Background Light Using Gallilean Satellites Eclip...

Observation	Proposal 2134, Observation 2: Europa eclipse									Wed Jun 24 16:00:09 GMT 2026
	Diagnostic Status: Warning									
Observing Template: NIRCam Imaging										
Diagnostics	(Europa eclipse (Obs 2)) Warning (Form): No dither offsets specified for observation									
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Solar System Targets	(Europa eclipse (Obs 2)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.									
	#	Name	Level 1			Level 2			Level 3	
Template	(2)	EUROPA-OFFSET2	STD=JUPITER			TYPE=POS_ANGLE,RAD=100,ANG=250,REF=NO RTH				
	Comments: Extended=YES									
Dithers	Module					Subarray				
	B					FULL				
Spectral Elements	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	NONE				SMALL-GRID-DITHER				1
Special Requirements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F140M	F356W	SHALLOW4	5	4	4	1	1062.94	
	2	F140M	F356W	SHALLOW4	5	4	4	1	1062.94	
	3	F140M	F356W	SHALLOW4	5	4	4	1	1062.94	
	4	F140M	F356W	SHALLOW4	5	4	4	1	1062.94	
Special Requirements	Between Dates 10-NOV-2025:12:30:00 and 10-NOV-2025:14:30:00									
	ECLIPSE UMBRAL FULL OF EUROPA BY JUPITER FROM JWST DEFAULT WINDOW: NOT OCCULTATION OF EUROPA-OFFSET2 BY JUPITER FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF EUROPA-OFFSET2 BY IO FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF EUROPA-OFFSET2 BY EUROPA FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF EUROPA-OFFSET2 BY GANYMEDE FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF EUROPA-OFFSET2 BY CALLISTO FROM JWST DEFAULT WINDOW: SEPARATION OF EUROPA-OFFSET2 IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA-OFFSET2 EUROPA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA-OFFSET2 GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF EUROPA-OFFSET2 CALLISTO FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE EUROPA-OFFSET2 FROM JWST LESS THAN 0.075 SEPARATION OF EUROPA JUPITER FROM JWST GREATER THAN 5"									

Proposal 2134 - Observation 3 - Absolute Brightness Measuremt of the Extragalactic Background Light Using Gallilean Satellites Eclip...

Observation	Proposal 2134, Observation 3: Ganymede eclipse									Wed Jun 24 16:00:09 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRCam Imaging									
Diagnostics	(Ganymede eclipse (Obs 3)) Warning (Form): No dither offsets specified for observation									
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Ganymede eclipse (Obs 3)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.									
Solar System Targets	#	Name	Level 1			Level 2		Level 3		
	(5)	GANYMEDE-OFFSET-V3	STD=JUPITER			TYPE=POS_ANGLE,RAD=90,ANG=234,REF=NOR		TH		
	Comments: Target for Obs 3. Extended=Unknown									
Template	Module					Subarray				
	B					FULL				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	NONE				SMALL-GRID-DITHER				1
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F140M	F356W	SHALLOW4	5	4	4	1	1062.94	
	2	F140M	F356W	SHALLOW4	5	4	4	1	1062.94	
	3	F140M	F356W	SHALLOW4	5	4	4	1	1062.94	
	4	F140M	F356W	SHALLOW4	5	4	4	1	1062.94	
	5	F140M	F356W	SHALLOW4	5	4	4	1	1062.94	
	6	F140M	F356W	SHALLOW4	5	4	4	1	1062.94	
Special Requirements	Aperture PA Range 242.05262691 to 242.05262691 Degrees (V3 242.0 to 242.0)									
	ECLIPSE UMBRAL FULL OF GANYMEDE BY JUPITER FROM JWST DEFAULT WINDOW: NOT OCCULTATION OF GANYMEDE-OFFSET-V3 BY JUPITER FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF GANYMEDE-OFFSET-V3 BY IO FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF GANYMEDE-OFFSET-V3 BY EUROPA FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF GANYMEDE-OFFSET-V3 BY GANYMEDE FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF GANYMEDE-OFFSET-V3 BY CALLISTO FROM JWST DEFAULT WINDOW: SEPARATION OF GANYMEDE-OFFSET-V3 IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF GANYMEDE-OFFSET-V3 EUROPA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF GANYMEDE-OFFSET-V3 GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF GANYMEDE-OFFSET-V3 CALLISTO FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE GANYMEDE-OFFSET-V3 FROM JWST LESS THAN 0.03									

Proposal 2134 - Observation 4 - Absolute Brightness Measuremt of the Extragalactic Background Light Using Gallilean Satellites Eclip...

Observation	Proposal 2134, Observation 4: Ganymede eclipse									Wed Jun 24 16:00:09 GMT 2026
	Diagnostic Status: Warning									
	Observing Template: NIRCam Imaging									
Diagnostics	(Ganymede eclipse (Obs 4)) Warning (Form): No dither offsets specified for observation									
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
	(Ganymede eclipse (Obs 4)) Informational (Form): The Visit Planner and Spike may produce different schedulability results.									
Solar System Targets	#	Name	Level 1			Level 2		Level 3		
	(6)	GANYMEDE-OFFSET-V4	STD=JUPITER			TYPE=POS_ANGLE,RAD=105,ANG=258,REF=NO RTH				
	Comments: Target for Observation 4 Extended=Unknown									
Template	Module					Subarray				
	B					FULL				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions
	1	NONE				SMALL-GRID-DITHER				1
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
	1	F140M	F356W	SHALLOW4	5	4	4	1	1062.94	
	2	F140M	F356W	SHALLOW4	5	4	4	1	1062.94	
	3	F140M	F356W	SHALLOW4	5	4	4	1	1062.94	
	4	F140M	F356W	SHALLOW4	5	4	4	1	1062.94	
	5	F140M	F356W	SHALLOW4	5	4	4	1	1062.94	
	6	F140M	F356W	SHALLOW4	5	4	4	1	1062.94	
Special Requirements	Aperture PA Range 295.05262691 to 295.05262691 Degrees (V3 295.0 to 295.0)									
	ECLIPSE UMBRAL FULL OF GANYMEDE BY JUPITER FROM JWST DEFAULT WINDOW: NOT OCCULTATION OF GANYMEDE-OFFSET-V4 BY JUPITER FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF GANYMEDE-OFFSET-V4 BY IO FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF GANYMEDE-OFFSET-V4 BY EUROPA FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF GANYMEDE-OFFSET-V4 BY GANYMEDE FROM JWST DEFAULT WINDOW: NOT ECLIPSE PENUMBRAL PARTIAL OF GANYMEDE-OFFSET-V4 BY CALLISTO FROM JWST DEFAULT WINDOW: SEPARATION OF GANYMEDE-OFFSET-V4 IO FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF GANYMEDE-OFFSET-V4 EUROPA FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF GANYMEDE-OFFSET-V4 GANYMEDE FROM JWST GREATER THAN 10" DEFAULT WINDOW: SEPARATION OF GANYMEDE-OFFSET-V4 CALLISTO FROM JWST GREATER THAN 10" DEFAULT WINDOW: ANGULAR RATE GANYMEDE-OFFSET-V4 FROM JWST LESS THAN 0.03 SEPARATION OF JUPITER GANYMEDE FROM JWST GREATER THAN 4"									